

CBSE Class 12 Physics — Electromagnetic Induction

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

Q1. The magnetic flux linked with a coil (in Weber) is given by the equation

$\phi = 5t^2 + 3t + 16$. The induced EMF in the coil at time $t = 4$ s will be: (a) -27 V (b) -43 V (c) -108 V (d) 210 V [PYQ 2021] Very High

Q2. A rectangular, a square, a circular, and an elliptical loop, all in the $x - y$ plane, are moving out of a uniform magnetic field with a constant velocity $\vec{v} = v\hat{i}$. The magnetic field is directed along the negative z-axis. The induced emf, during the passage of these loops out of the field region, will NOT remain constant for: (a) any of the four loops (b) the circular and elliptical loops (c) the rectangular, circular and elliptical loops (d) only the elliptical loops [PYQ 2022] Very High

Q3. Two coils are placed close to each other. The mutual inductance of the pair of coils strictly depends upon the: (a) rate at which current changes in the two coils (b) relative position and orientation of the coils (c) rate at which voltage is induced across the two coils (d) currents in the two coils [PYQ 2021] High

Q4. A constant current is flowing through a solenoid. An iron rod is inserted into the solenoid along its axis. Which of the following quantities will NOT increase? (a) The magnetic field at the centre (b) The magnetic flux linked with the solenoid (c) The rate of Joule heating (d) The self-inductance of the solenoid [PYQ 2021] High

Q5. The self-inductance of a solenoid of 600 turns is 108 mH. The self-inductance of a coil having 500 turns with the same length, same radius, and same core medium will be: (a) 95 mH (b) 90 mH (c) 85 mH (d) 75 mH [PYQ 2021] Very High

Q6. The magnetic flux linked with a coil changes by 2×10^{-2} Wb when the current through it changes by 0.01 A. The self-inductance of the coil is: (a) 2 H (b) 0.5 H (c) 10^{-4} H (d) 200 H [PYQ 2020] Very High

Q7. A coil of area 100 cm^2 is kept with its plane making an angle of 30° with a magnetic field of 10^{-1} T. The magnetic field is reduced to zero in 10^{-4} s. The induced emf in the coil is: (a) $5\sqrt{3}$ V (b) $50\sqrt{3}$ V (c) 5.0 V (d) 50.0 V [PYQ 2021] High

Q8. A conducting square loop of side L and resistance R moves in its plane with uniform velocity v perpendicular to one of its sides. A uniform magnetic field B , constant in time and space, pointing perpendicular and into the plane of the loop exists everywhere. The current induced in the loop is: (a) BLv/R clockwise (b) BLv/R anticlockwise (c) $2BLv/R$ anticlockwise (d) Zero [PYQ 2021] Very High

Q9. A long straight current-carrying wire normally passes through the exact center of a circular loop. If the current through the wire increases, will there be an induced emf in the

loop? [PYQ 2017] High

Q10. Predict the polarity of the capacitor in a circuit where a circular loop is connected to a capacitor, and the North pole of a bar magnet is moved towards the loop. [PYQ 2022] Very High

Q11. The electric current flowing in a straight horizontal wire from point B to point A is decreasing. Find the direction of the induced current in a metallic circular loop kept exactly above the wire. [PYQ 2014] High

Q12. A flexible wire of irregular shape turns into a circular shape when placed in a region of magnetic field which is directed normal to the plane of the loop away from the reader. Predict the direction of the induced current in the wire. [PYQ 2014] Medium

Q13. A solenoid with N loops of wire tightly wrapped around an iron-core carries an electric current I . If the current through this solenoid is reduced to half, what change would you expect in the inductance L of the solenoid? [PYQ 2021] Very High

Q14. The number of turns of a solenoid is strictly doubled without changing its length and area of cross-section. The self-inductance of the solenoid will become __ times. [PYQ 2020] Very High

Q15. A conducting rod of length l is kept strictly parallel to a uniform magnetic field $\vec{B}$. It is moved along the magnetic field with a velocity $\vec{v}$. What is the value of the emf induced in the conductor? [CBSE 2020] High

Q16. What is the SI unit of magnetic flux? Write its corresponding dimensional formula. [Practice] Medium

Q17. Write the mathematical expression defining 1 Henry of mutual inductance. [Practice] High

Q18. Assertion (A): The induced emf in a coil always opposes the change in magnetic flux that produces it. **Reason (R):** This is in strict accordance with the law of conservation of energy. [Practice] Very High

Q19. Assertion (A): An artificial satellite with a metal surface is moving about the Earth in a circular orbit. An induced current will flow in it. **Reason (R):** As the satellite moves, it continuously cuts the magnetic flux of the Earth's magnetic field. [Practice] Medium

Q20. Assertion (A): The mutual inductance of a given pair of coils depends on their geometric separation. **Reason (R):** As the distance between the two coils increases, the magnetic flux linked with the secondary coil due to the primary coil drastically decreases. [Practice] Very High

2-Mark Questions

Q21. State Faraday's laws of electromagnetic induction. Write its mathematical expression. [Practice] Very High

Q22. State Lenz's law. Give one conceptual example to illustrate this law. Briefly justify how Lenz's law is a consequence of the principle of conservation of energy. [PYQ 2014] Very High

Q23. How does the mutual inductance of a pair of coils specifically change when: (A) the geometric distance between the coils is decreased, and (B) the number of turns in one of the coils is decreased? [PYQ 2023] Very High

Q24. Plot a clear graph showing the variation of induced emf (ϵ) with the rate of change of current (dI/dt) flowing through an ideal inductor. What does the slope of this graph represent? [PYQ 2020] High

Q25. A rectangular loop and a circular loop are moving out of a uniform magnetic field region to a field-free region with a constant velocity v . The field is entirely normal to the loops. In which loop do you expect the induced emf to remain constant during the passage out of the field region? Justify your answer. [PYQ 2022] Very High

Q26. A pair of adjacent coils has a mutual inductance of 1.5 H . If the current in one coil drops from 20 A to zero in 0.5 s , what is the change of magnetic flux linked with the other coil? [PYQ 2022] High

Q27. An inductor L of reactance X_L is connected in series with a glowing bulb B and an AC source. How would the brightness of the bulb physically change when an iron rod is completely inserted into the inductor? Justify your answer. [PYQ 2015] High

Q28. The current flowing in two coils of self-inductances $L_1 = 16 \text{ mH}$ and $L_2 = 12 \text{ mH}$ are increasing at the exact same rate. If the instantaneous power supplied to the two coils is equal, find the ratio of (A) the induced voltages across them, and (B) the currents flowing in them. [PYQ 2014] Very High

Q29. The current flowing through an inductor of self-inductance L is continuously increasing. Plot a graph showing the variation of (A) Magnetic flux versus the current, and (B) Magnetic potential energy stored versus the current. [PYQ 2014] High

Q30. Define self-inductance of a coil. Obtain the mathematical expression for the magnetic energy stored in an inductor L to build up a current I through it. [PYQ 2017] Very High

Q31. A long solenoid with 15 turns per cm has a small loop of area 2.0 cm^2 placed exactly inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 2.0 A to 4.0 A in 0.1 s , what is the induced emf in the inner loop while the current is changing? [PYQ 2016] High

Q32. A metallic rod PQ of length l is rotated with an angular velocity ω about an axis passing through its mid-point (O) and perpendicular to the plane of the paper, in a uniform magnetic field B . What is the potential difference developed between the two ends of the rod, P and Q? [PYQ 2020] Medium

3-Mark Questions

Q33. Using the concept of the Lorentz magnetic force on free charge carriers, derive the expression for the motional emf ($\epsilon = Blv$) induced across the ends of a straight conducting

rod of length l moving with a constant velocity v perpendicular to a uniform magnetic field B . [Practice] Very High

Q34. A metallic rod of length l is rotated with a frequency ν , with one end hinged at the centre and the other end sweeping the circumference of a circular metallic ring of radius l . A constant and uniform magnetic field B parallel to the axis is present everywhere. Derive the expression for the emf induced between the centre and the metallic ring. [PYQ 2017] Very High

Q35. Define the term mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of the same length wound one over the other. [PYQ 2017] Very High

Q36. Two concentric circular coils, one of small radius r and the other of large radius R , such that $R \gg r$, are placed coaxially with their centres coinciding. Obtain the analytical expression for the mutual inductance of the arrangement. [PYQ 2016] High

Q37. A rectangular loop PQMN with a movable arm PQ of length **10 cm** and resistance **2 Ω** is placed in a uniform magnetic field of **0.1 T** acting perpendicular to the plane of the loop. The resistances of the other arms are negligible. Calculate: (A) the emf induced in the arm PQ, and (B) the current induced in the loop when the arm PQ is moved with a velocity of **20 m/s**. [PYQ 2014] Very High

Q38. (A) Draw a schematic labelled diagram of an AC generator. (B) Explain its working principle and obtain the expression for the instantaneous value of the emf induced in the rotating coil. [PYQ 2019] Very High

Q39. The current through two identical inductors, of self-inductances **12 mH** and **30 mH** respectively, is increasing with time at the exact same rate. (A) Draw a graph showing the variation of the emf induced with the rate of change of current in each inductor. (B) Compare the instantaneous energy stored in the coils if the power actively dissipated is identical. [PYQ 2017] High

Q40. A conducting rod PQ of length **20 cm** and resistance **0.1 Ω** rests on two smooth parallel rails of negligible resistance. It can slide freely and the arrangement is positioned between the poles of a permanent magnet producing a uniform magnetic field $B = 0.4 \text{ T}$. If the ends of the rails are short-circuited, find: (A) the external force required to move the rod with a uniform velocity $v = 10 \text{ cm/s}$, and (B) the mechanical power required to do so. [PYQ 2020] High

Q41. When a conducting loop of resistance **10 Ω** and area **10 cm²** is abruptly removed from an external magnetic field acting normally, the variation of induced current I with time t is a straight line dropping from a maximum to zero in a short time Δt . If the total charge passed through the loop is Q , deduce an expression for the magnitude of the applied magnetic field. [PYQ 2020] Medium

Q42. Derive an expression for the self-inductance of a long air-cored solenoid of length l , cross-sectional area A , and having N total number of turns. [PYQ 2014] High

5-Mark Questions (Long Answer/Case-Study)

Q43. (A) Define mutual inductance. (B) Consider two concentric circular loops of radii **1 cm** and **20 cm** placed coaxially with their centers coinciding. Find the mutual inductance of the arrangement. (C) If the current passed through the outer loop is changed at a constant rate of **5 A/ms**, calculate the magnitude of the emf induced in the inner loop. Assume the magnetic field on the inner loop is uniform across its small area. [PYQ 2020] Very High

Q44. Figure shows a rectangular conductor PQRS in which the arm PQ is free to move in a uniform magnetic field ***B*** pointing perpendicular to the plane of the paper. The field extends exclusively from ***x* = 0** to ***x* = *b*** and is identically zero for ***x* > *b***. Assume that only the arm PQ possesses resistance ***r***. (A) When the arm PQ is pulled outward from ***x* = 0** to ***x* = 2*b*** and then moved backward to ***x* = 0** with a constant speed ***v***, obtain the mathematical expressions for the magnetic flux ***ϕ*** and the induced emf ***ε***. (B) Sketch the graphical variations of flux ***ϕ*** and induced emf ***ε*** with distance **0 < *x* < 2*b***. [PYQ 2023] Very High

Q45. (A) Describe a simple experimental activity to explicitly demonstrate that the polarity of an emf induced in a coil is always such that it tends to produce a current which opposes the change in magnetic flux that produces it (Lenz's Law). (B) What are eddy currents? How are they fundamentally produced? (C) Give two practical applications of eddy currents in daily life or industry. [PYQ 2014 / Practice] High

Q46. (A) Define the self-inductance of a coil and state its SI unit. (B) An iron-core solenoid has a self-inductance of **2.8 H**. When the iron core is entirely removed from the solenoid, the self-inductance steeply drops to **2 mH**. Calculate the relative magnetic permeability (***μ_r***) of the iron core used. (C) Plot a graph showing the variation of induced emf across an inductor with the rate of change of current ***dI/dt***. [PYQ 2017 / Practice] Very High

Q47. (A) Draw a schematic labelled diagram of an AC generator. State its underlying working principle. (B) A circular coil of radius **10 cm** and 20 turns is rotated about its vertical diameter with an angular speed of **50 rad/s** in a uniform horizontal magnetic field of **3.0×10^{-2} T**. Calculate the maximum emf and the average emf induced in the coil over one complete revolution. (C) If the coil forms a closed loop of total resistance **10 Ω**, calculate the maximum current in the coil and the average power loss strictly due to Joule heating. [PYQ 2019] Very High

Q48. (A) State Faraday's law of electromagnetic induction. (B) A rectangular loop of length ***l*** and breadth ***b*** is mechanically pulled out of a uniform magnetic field ***B*** perpendicular to its plane with a constant velocity ***v***. Derive the expression for the mechanical force required to pull the loop. (C) Show mathematically that the electrical power dissipated as Joule heat in the loop is exactly equal to the mechanical power expended in pulling the loop. [Practice] High

Q49. Case Study: Eddy Currents and Damping When thick metal plates or bulk pieces of conductors are moved through a changing magnetic field, circulating currents are induced within the body of the metal. These are called eddy currents. Due to the very low resistance of bulk metals, eddy currents can be quite large, leading to significant Joule heating and causing strong electromagnetic damping. (a) How can eddy currents be effectively minimized

in the heavy core of a transformer? (b) Name one modern application where the heating effect of eddy currents is purposely utilized. (c) A flat copper plate is allowed to swing freely like a pendulum between the poles of a strong electromagnet. What happens to its oscillatory motion when the electromagnet is switched on, and why? (d) If the same copper plate has deep slots cut into it, how is the electromagnetic damping affected compared to the solid plate? [Practice] Very High

Q50. Case Study: Motional EMF in Aeronautics An airplane with a metallic body acts as a moving conductor in the Earth's magnetic field. As it flies horizontally, its wings cut across the magnetic field lines of the Earth, generating a motional electromotive force (emf) between the tips of the wings. An airplane is travelling horizontally at a speed of **900 km/h** in a region where the Earth's total magnetic field is $5 \times 10^{-5} \text{ T}$ and the angle of dip is 30° . The wingspan (distance between wingtips) of the plane is **20 m**. (a) Which specific component of the Earth's magnetic field (horizontal or vertical) induces an emf across the wings of the horizontally flying plane? (b) Calculate the magnitude of the relevant vertical component of the Earth's magnetic field in this region. (c) Calculate the exact motional emf induced across the ends of the wings. (d) Does the polarity of the induced emf depend on whether the airplane is flying East-to-West or West-to-East? Briefly justify. [Practice] Very High

Answer Key

Q1: (b) -43 V . ($\epsilon = -d\phi/dt = -(10t + 3)$. At $t = 4$, $\epsilon = -(40 + 3) = -43 \text{ V}$). **Q2:** (b) the circular and elliptical loops. (For these, the length cutting the magnetic field lines changes continuously, so motional emf Blv is not constant). **Q3:** (b) relative position and orientation of the coils. **Q4:** (c) The rate of Joule heating. (Heating I^2R only depends on current and resistance, which remain constant). **Q5:** (d) **75 mH**. ($L \propto N^2$. $L_2 = 108 \times (500/600)^2 = 108 \times 25/36 = 75 \text{ mH}$). **Q6:** (a) **2 H**. ($L = \Delta\phi/\Delta I = 2 \times 10^{-2}/0.01 = 2 \text{ H}$). **Q7:** (c) **5.0 V**. (Angle with normal $\theta = 90^\circ - 30^\circ = 60^\circ$. $\Delta\phi = BA \cos(60^\circ) = 10^{-1} \times 100 \times 10^{-4} \times 0.5 = 5 \times 10^{-4} \text{ Wb}$. $\epsilon = \Delta\phi/\Delta t = 5 \times 10^{-4}/10^{-4} = 5 \text{ V}$). **Q8:** (d) Zero. (Since the loop moves completely inside a uniform magnetic field, the flux linked remains constant, so $d\phi/dt = 0$). **Q9:** No. (Magnetic field lines form concentric circles in the plane of the loop. Flux $\phi = \vec{B} \cdot \vec{A} = 0$, so $\epsilon = 0$). **Q10:** The face of the loop towards the magnet develops a North polarity to oppose the magnet. Hence, current is anticlockwise. The plate connected to the arrow tip of the anticlockwise current becomes positive. **Q11:** Current A to B is decreasing, causing inward magnetic flux to decrease. By Lenz's law, induced current opposes this by creating inward flux, so it flows clockwise in the loop. **Q12:** The irregular wire expands to form a circle (increasing area). To oppose the increasing outward flux, the induced current produces an inward magnetic field, so it flows clockwise. **Q13:** Inductance L remains completely unchanged (it depends only on the geometry and core material of the solenoid, not the current). **Q14:** Four (4). ($L \propto N^2$. If N is doubled, N^2 becomes 4). **Q15:** Zero. (Since $\vec{v}$ is parallel to $\vec{B}$, the angle is zero, motional emf $\epsilon = l \cdot (\vec{v} \times \vec{B}) = 0$). **Q16:** Weber (Wb).

Dimensional formula: $[ML^2T^{-2}A^{-1}]$. **Q17:** Mutual inductance is 1 Henry if a current change of 1 A/s in the primary coil induces an EMF of 1 V in the secondary coil. ($M = |\epsilon|/(dI/dt)$). **Q18:** (a) Both A and R are true, and R is the correct explanation of A. **Q19:** (a) Both A and R are true, and R is the correct explanation of A. **Q20:** (a) Both A and R are true, and R is the correct explanation of A. **Q21:** The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux linked with the circuit. $\epsilon = -d\phi_B/dt$. **Q22:** Lenz's law states that the direction of induced current is such that it opposes the change in magnetic flux that produces it. If we push a magnet's N-pole into a coil, the coil acts as an N-pole. We must do mechanical work against this repulsion, which transforms into electrical energy, proving conservation of energy. **Q23:** (A) Increases (flux linkage increases). (B) Decreases ($M \propto N_1N_2$). **Q24:** A straight line passing through the origin. The slope represents the self-inductance L of the inductor ($\epsilon = L \cdot dI/dt$). **Q25:** Rectangular loop. For a rectangular loop, the length l cutting the field is constant. For a circular loop, the vertical chord length cutting the field changes, so ϵ is not constant. **Q26:** $\Delta\phi = M\Delta I = 1.5 \times (20 - 0) = 30 \text{ Wb}$. **Q27:** Brightness decreases. Inserting the iron rod heavily increases the self-inductance L , which increases inductive reactance X_L . The total impedance increases, reducing the current and hence the brightness. **Q28:** (A) $\epsilon = L(dI/dt)$. Ratio $\epsilon_1/\epsilon_2 = L_1/L_2 = 16/12 = 4/3$. (B) Power $P = \epsilon I$. Since $P_1 = P_2$, $\epsilon_1 I_1 = \epsilon_2 I_2 \Rightarrow I_1/I_2 = \epsilon_2/\epsilon_1 = 3/4$. **Q29:** (A) ϕ vs I is a straight line through origin (slope L). (B) U vs I is an upward opening parabola ($U = \frac{1}{2}LI^2$). **Q30:** Self-inductance is the property of a coil by virtue of which it opposes any change in the current flowing through it. Work $dW = \epsilon I dt = L \frac{dI}{dt} I dt = LI dI$. Integrating gives $U = \frac{1}{2}LI^2$. **Q31:** $B = \mu_0 n I$. $n = 1500 \text{ turns/m}$. $\Delta B = \mu_0 n \Delta I = (4\pi \times 10^{-7}) \times 1500 \times 2.0 = 12\pi \times 10^{-4} \text{ T}$. Induced emf $\epsilon = A \frac{\Delta B}{\Delta t} = (2 \times 10^{-4}) \times \frac{12\pi \times 10^{-4}}{0.1} = 24\pi \times 10^{-7} \text{ V} \approx 7.54 \times 10^{-6} \text{ V}$. **Q32:** Zero. The emf induced in the left half (O to P) is exactly equal and opposite to the emf induced in the right half (O to Q). Thus $V_P - V_Q = 0$. **Q33:** A charge q in the rod experiences Lorentz force $F = qvB$. This drives charges to the ends, building an electric field E until $qE = qvB \Rightarrow E = vB$. The potential difference is $\epsilon = El = vBl$. **Q34:** Consider a small element dr at distance r . Motional emf $d\epsilon = Bvdr = B(r\omega)dr$. Integrating from 0 to l : $\epsilon = \int_0^l B\omega r dr = \frac{1}{2}B\omega l^2$. Since $\omega = 2\pi\nu$, $\epsilon = B\pi\nu l^2$. **Q35:** Flux linkage in coil 2 due to current in coil 1: $\phi_2 = MI_1$. Solenoid 1 field $B_1 = \mu_0 n_1 I_1$. Flux $\phi_2 = N_2 B_1 A = (n_2 l)(\mu_0 n_1 I_1)A$. $M = \mu_0 n_1 n_2 Al$. **Q36:** Field at center of large coil $B = \frac{\mu_0 I}{2R}$. Flux linked with small coil $\phi = B \times (\pi r^2) = \frac{\mu_0 \pi r^2}{2R} I$. Thus, $M = \frac{\mu_0 \pi r^2}{2R}$. **Q37:** (A) $\epsilon = Blv = 0.1 \times 0.1 \times 20 = 0.2 \text{ V}$. (B) Current $I = \epsilon/R = 0.2/2 = 0.1 \text{ A}$. **Q38:** (A) Diagram showing armature, magnets, slip rings, brushes. (B) Principle: EMI. $\phi = NBA \cos(\omega t)$. $\epsilon = -d\phi/dt = NBA\omega \sin(\omega t)$. **Q39:** (A) Graphs are straight lines through origin; steeper slope for 30 mH . (B) $U_1/U_2 = (L_1/L_2)(I_1/I_2)^2$. Since P is same, $I_1/I_2 = \epsilon_2/\epsilon_1 = L_2/L_1$. Thus $U_1/U_2 = (L_1/L_2)(L_2/L_1)^2 = L_2/L_1 = 30/12 = 2.5$. **Q40:** (A) $\epsilon = Blv = 0.4 \times 0.2 \times 0.1 = 0.008 \text{ V}$. $I = \epsilon/R = 0.08 \text{ A}$. Force $F = BIl = 0.4 \times 0.08 \times 0.2 = 0.0064 \text{ N}$. (B) Power $P = Fv = 0.0064 \times 0.1 = 0.00064 \text{ W}$ (or $6.4 \times 10^{-4} \text{ W}$). **Q41:** Total charge $Q = \int I dt = \int \frac{1}{R} \frac{d\phi}{dt} dt = \frac{\Delta\phi}{R} = \frac{BA}{R}$. Therefore, $B = \frac{QR}{A}$. **Q42:** $B = \mu_0 \frac{N}{l} I$. Flux linkage

$N\phi = N(BA) = N(\mu_0 \frac{N}{l} I)A$. Since $N\phi = LI$, $L = \frac{\mu_0 N^2 A}{l}$. **Q43:** (A) Mutual inductance is the property by which an EMF is induced in one coil due to a changing current in a neighboring coil. (B) $M = \frac{\mu_0 \pi r^2}{2R} = \frac{4\pi \times 10^{-7} \times \pi \times 10^{-4}}{2 \times 0.20} = \pi^2 \times 10^{-10} \text{ H}$. (C) $\epsilon = M \frac{dI}{dt} = (\pi^2 \times 10^{-10}) \times 5000 = 5\pi^2 \times 10^{-7} \text{ V}$. **Q44:** (A) $0 < x < b$: $\phi = Blx$, $\epsilon = -Blv$. $b < x < 2b$: $\phi = Blb$ (constant), $\epsilon = 0$. Backward $2b < x < b$: $\phi = Blb$, $\epsilon = 0$. $b < x < 0$: ϕ decreases to 0, $\epsilon = +Blv$. (B) ϕ graph rises linearly to b , stays flat to $2b$ and back to b , drops linearly to 0. ϵ graph is a negative square pulse for 0 to b , zero everywhere else until b to 0 on return where it is a positive square pulse. **Q45:** (A) Drop a strong magnet down a conducting copper pipe; it falls slower than g due to opposing induced currents, proving Lenz's law. (B) Circulating currents induced in bulk conductors when exposed to changing magnetic flux. (C) Electromagnetic braking in trains, Induction furnaces. **Q46:** (A) Self-inductance is the ratio of magnetic flux linkage to the current in the coil. SI unit: Henry. (B) $L = \mu_r L_0$. $\mu_r = L/L_0 = 2.8/(2 \times 10^{-3}) = 1400$. (C) Straight line passing through the origin. **Q47:** (A) Diagram showing coils, field, slip rings. Principle: EMI (changing flux induces EMF). (B) Maximum EMF $\epsilon_0 = NBA\omega = 20 \times (3.0 \times 10^{-2}) \times (\pi \times 0.1^2) \times 50 = 3\pi \times 10^{-1} = 0.942 \text{ V}$. Average EMF over a complete cycle is zero. (C) $I_{max} = \epsilon_0/R = 0.942/10 = 0.0942 \text{ A}$. $P_{avg} = \frac{\epsilon_0 I_{max}}{2} = \frac{0.942 \times 0.0942}{2} = 0.044 \text{ W}$. **Q48:** (A) Induced emf equals the negative rate of change of magnetic flux. (B) $\epsilon = Blv \Rightarrow I = Blv/R$. Force $F = BIl = B(Blv/R)l = \frac{B^2 l^2 v}{R}$. (C) Mechanical Power $P_{mech} = Fv = \frac{B^2 l^2 v^2}{R}$. Electrical power $P_{elec} = I^2 R = (\frac{Blv}{R})^2 R = \frac{B^2 l^2 v^2}{R}$. Thus $P_{mech} = P_{elec}$. **Q49:** (a) By using a laminated core instead of a solid block (thin sheets insulated from each other). (b) Induction furnaces for melting metals. (c) The swing gets strongly damped and stops quickly because eddy currents are generated, causing opposing magnetic forces. (d) Slots heavily disrupt the circulating paths of eddy currents, reducing them significantly and thus reducing the damping effect. **Q50:** (a) The vertical component of the Earth's magnetic field (B_v). (b) $B_v = B_{total} \sin(Dip) = 5 \times 10^{-5} \times \sin(30^\circ) = 2.5 \times 10^{-5} \text{ T}$. (c) Velocity $v = 900 \times (5/18) = 250 \text{ m/s}$. $\epsilon = B_v l v = (2.5 \times 10^{-5}) \times 20 \times 250 = 0.125 \text{ V}$. (d) Yes. Reversing the flight direction (velocity vector) reverses the direction of $\vec{v} \times \vec{B}$, flipping the polarity of the induced motional emf.